

EX 5 – Demonstrate transformation, scaling, binning fixing skewed values and sampling.

AIM :

To demonstrate data transformation, scaling, binning, handling of skewed data, and sampling techniques using a dataset in R/Python.

ALGORITHM :

1. Data Transformation

1. Import required libraries (Pandas, NumPy)
2. Load the dataset
3. Apply transformations such as:
 - Log transformation
 - Square root transformation
 - Encoding categorical variables (if needed)

2. Scaling

1. Select numerical features
2. Apply scaling methods:
 - Min-Max Scaling (values between 0 and 1)
 - Standardization (mean = 0, standard deviation = 1)
3. Replace original values with scaled values

3. Binning

1. Select a continuous variable
2. Divide data into intervals (bins)
3. Assign labels to each bin (e.g., low, medium, high)
4. Convert continuous data into categorical data

4. Fixing Skewed Values

1. Identify skewness using histogram or skew function
2. Apply transformations:

- Log transformation
 - Box-Cox transformation
3. Recheck distribution after transformation

5. Sampling

1. Choose sampling method:
 - Random sampling
 - Stratified sampling
2. Select a subset of data from the dataset
3. Ensure sample represents the population

CODE :

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler

df = pd.read_csv("StudentsPerformance.csv")
df["math_log"] = np.log1p(df["math score"])

scaler = MinMaxScaler()
df[["math_scaled","reading_scaled","writing_scaled"]] = scaler.fit_transform(
    df[["math score","reading score","writing score"]]
)

df["math_bin"] = pd.cut(df["math score"], bins=5, labels=["Very Low","Low","Medium","High","Very High"])
sample_df = df.sample(n=50)

print(df.head())
print(sample_df.head())
```

OUTPUT :

```

gender race/ethnicity parental level of education lunch \
0 female group B bachelor's degree standard \
1 female group C some college standard
2 female group B master's degree standard
3 male group A associate's degree free/reduced
4 male group C some college standard

test preparation course math score reading score writing score math_log \
0 none 72 72 74 4.290459
1 completed 69 90 88 4.248495
2 none 90 95 93 4.510860
3 none 47 57 44 3.871201
4 none 76 78 75 4.343805

math_scaled reading_scaled writing_scaled math_bin
0 0.72 0.662651 0.711111 High
1 0.69 0.879518 0.866667 High
2 0.90 0.939759 0.922222 Very High
3 0.47 0.481928 0.377778 Medium
4 0.76 0.734940 0.722222 High

gender race/ethnicity parental level of education lunch \
809 male group B bachelor's degree standard
32 female group E master's degree free/reduced
408 female group D high school free/reduced
284 female group B some high school standard
338 female group B some high school free/reduced

test preparation course math score reading score writing score \
809 none 59 54 51
32 none 56 72 65
408 completed 52 57 56
284 none 37 46 46
338 none 24 38 27

math_log math_scaled reading_scaled writing_scaled math_bin
809 4.094345 0.59 0.445783 0.455556 Medium
32 4.043051 0.56 0.662651 0.611111 Medium
408 3.970292 0.52 0.481928 0.511111 Medium
284 3.637586 0.37 0.349398 0.400000 Low
338 3.218876 0.24 0.253012 0.188889 Low

```

RESULT :

Thus the above program is executed and the result is obtained.